

Software Test Plan (STP)

Project: Online Bookstore

Version: 1.0

Authors:

PES2202400289 – Zia Kadijah (SRN: PES2UG24CS620)
PES2202402365 – Hithesh Nanjunda Swamy (SRN: PES2UG24CS635)
PES2202400125 – Nidhi Yedida (SRN: PES2UG24CS673)
PES2202402343 – Vidya Sri Ganesh (SRN: PES2UG24CS667)

Date: 15-09-2026

Status: Draft

1. Introduction

Purpose

This document defines the test plan for the Online Bookstore system v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope

Testing covers Online Bookstore features such as authentication, catalog browsing and search, cart and wishlist, checkout and payment, order tracking, seller/admin management, and reviews and ratings. Third-party payment gateway internals and shipping-carrier internal systems are excluded.

References

Online Bookstore SRS v1.0, Design Specifications v1.0, Web Security Standards, PCI-DSS.

Definitions

SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), API (Application Programming Interface), JWT (JSON Web Token), ORM (Object-Relational Mapping), CI/CD (Continuous Integration/Continuous Deployment), PCI-DSS (Payment Card Industry Data Security Standard).

2. Test Items

- Authentication & Account Management module
- Catalog Browsing & Search module
- Shopping Cart & Wishlist module
- Checkout & Payment module
- Order Management & Tracking module
- Seller & Admin Management module
- Reviews & Ratings module

3. Features to be Tested

Features mapped to SRS requirement IDs:

- BOOK-F-001: Register with email verification
- BOOK-F-002: Authenticate via JWT session token

- BOOK-F-010: Display paginated book catalog
- BOOK-F-011: Search and filter catalog
- BOOK-F-030 to BOOK-F-035: Checkout and payment processing
- BOOK-F-040 to BOOK-F-044: Order tracking and history
- BOOK-NF-001: Response time $\leq 2s$
- BOOK-NF-002: 99.5% availability
- BOOK-SR-003: No card data stored on application servers

4. Features Not to be Tested

- Payment gateway internal processing (vendor responsibility)
- Shipping/logistics partner internal systems
- Django admin framework internals (assumed tested upstream by the framework maintainers)

5. Test Approach / Strategy

Levels

- Unit tests (module-level Django tests)
- Integration tests (API $\leftrightarrow$ Database, API $\leftrightarrow$ payment gateway sandbox)
- System tests (end-to-end Online Bookstore user flows)
- Acceptance tests (UAT)

Types

- Functional testing (core features)
- Regression testing
- Performance testing (response time, load)
- Usability testing (UI clarity, accessibility)
- Security testing

Entry Criteria: Stable build deployed to staging, test data seeded, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password hashing (bcrypt/argon2) and confirm no plaintext storage
- TLS 1.2+ verification across all endpoints
- PCI-DSS compliance checks for the payment flow
- Fuzzing for input fields (search, forms, checkout)
- Penetration testing of authentication and RBAC flows

6. Test Environment

Hardware: Standard client devices (desktop, laptop, mobile) with internet connectivity; no dedicated hardware required.

Software: Django application (staging), PostgreSQL database, payment gateway sandbox (Razorpay/Stripe test mode).

Tools: Selenium/Playwright (UI automation), Postman (API testing), JMeter/Locust (performance), Jira (defect tracking), pytest/Django test framework (unit tests).

Test Data: Dummy buyer, seller, and admin accounts; sample book catalog; test payment cards.

7. Test Schedule

Milestones:

- Test case design: 18-Sep-2026
- Environment setup: 20-Sep-2026
- Test execution start: 22-Sep-2026
- Test execution end: 03-Oct-2026
- UAT: 05-Oct-2026 to 08-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Zia Kadijah	Prepare plan, coordinate execution
Test Engineer	Hithesh Nanjunda Swamy	Design & execute test cases, log defects
Developer	Nidhi Yedida	Support defect fixes and triage
Product Owner	Vidya Sri Ganesh	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the dev team
Test environment downtime	Maintain a backup environment on a cloud VM
Dependency on third-party payment gateway sandbox	Engage the gateway sandbox early and maintain a mock/stub for offline testing

11. Assumptions & Dependencies

- The payment gateway sandbox will be stable and available throughout testing
- Test data (accounts, catalog) will be provided before execution begins
- The staging environment mirrors the production configuration

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for >4 hours
- Build is too unstable (blocks >30% of test cases)

Resume testing if:

- Blocking defects are resolved
- Environment is stabilized

13. Test Case Management & Traceability

The RTM ensures mapping of SRS requirements to test cases. Example:

- BOOK-F-001 (Registration) → TC-Auth-01
- BOOK-F-002 (Authentication) → TC-Auth-02
- BOOK-F-010 (Catalog display) → TC-Cat-01
- BOOK-F-011 (Search & filter) → TC-Cat-02
- BOOK-NF-001 (Response time) → TC-Perf-01
- BOOK-SR-003 (No card data stored) → TC-Sec-04

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead		
Dev Lead		
Product Owner		